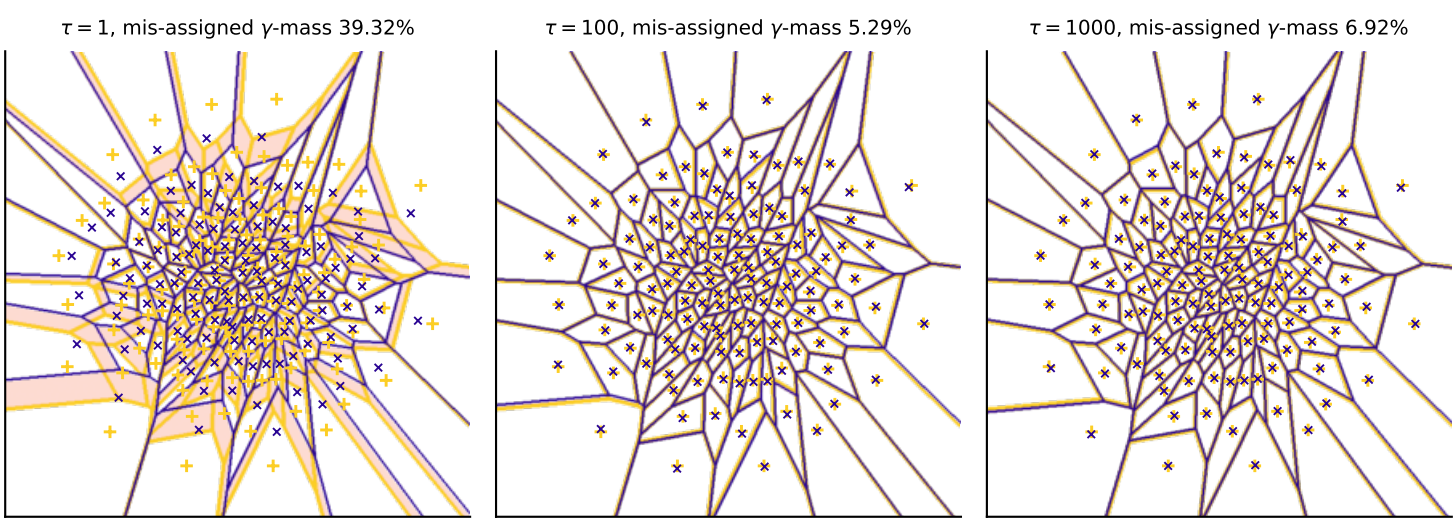

$\tau = 1$, mis-assigned γ-mass 39.32%
$\tau = 100$, mis-assigned γ-mass 5.29%
$\tau = 1000$, mis-assigned γ-mass 6.92%